

SEQUENTIAL CANDIDATE SELECTION FOR SCORE FOLLOWING IN SHEET MUSIC IMAGES

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ABSTRACT

Score following aligns a live performance to its score in real time, committing to a position at every instant with no lookahead. It concerns two modalities, image and audio, and the established response is to convert one into the other and align within a single modality, by optical music recognition or by transcription, each detour inheriting the errors of its conversion. Sheet-image score following conditions a visual detector on the audio, and what limits it is annotation, since alignments to notehead coordinates are costly enough that nearly all training data is synthesized and these systems lose much of their accuracy on a real instrument in a real room. Every remedy proposed for that gap has treated it as a failure of perception, yet we show that the detector already proposes a position within the standard tolerance at every scored onset of a real recording and that under the same realtime commitments the candidate set leaves only a small irreducible error, which places most of that loss in selection. We therefore recast the task as sequential selection over the candidates it already emits, combining a non-learned transition model with a set-based scorer that judges each candidate against its competitors. This exceeds the best reported result on real piano recordings while requiring neither test-time layout annotation nor detector retraining, and shows that the decision needs its own temporal history rather than a better image representation.

Index Terms— Score following, sheet music, audio-to-score alignment, realtime, music information retrieval

1. INTRODUCTION

Music alignment puts a performance and its score in correspondence, mapping each moment of audio to a position in the written music. The regimes differ in whether alignment may look ahead. Offline, known as audio-to-score alignment or music synchronisation, the whole recording is in hand, dynamic time warping settles every position at once [1], repeats and jumps are recoverable after the fact [2, 3], and the correspondences support cross-modal retrieval and piece identification [4].

In the online version, however, the system observes only what has been played, commits immediately and never revises.

This is score following [5], also called music tracking [6], in which the audio up to frame t fixes an estimate of the performer’s position that must be produced before frame $t + 1$ arrives. Automatic page turning [7], live computer accompaniment [8, 9] and score-synchronised display depend on it, and every result below, the oracles included, respects the constraint. The literature bridges the two modalities in three ways, differing in which side, if either, is converted.

Using optical music recognition, the first category renders the page as symbols, after which a classical follower tracks a discrete sequence of states, a line running from online dynamic time warping [1] through duration models [10], hidden semi-Markov models [11], particle filters [12], switching state-space models [13] and multi-agent tracking [6]. The second category converts the audio instead, pairing realtime piano transcription [14] with symbol-level tracking to strong effect [15], at the price of bounding the follower by the transcriber.

The third category learns the audio-to-pixel correspondence directly. Dorfer et al. [16] classified position into staff-level buckets on crops; reinforcement learning framed tracking as an agent on a score strip [17, 18]; CUNet [19] emitted an audio-conditioned segmentation of the page; and CYOLO [20] recast it as audio-conditioned object detection, CYOLO-SB [21] adding heads for the enclosing system and bar. CODA [22] cascades system, bar and note selection over layout regions with beam search and learned transition penalties and is the strongest published system here, though it reads the page’s system and bar boxes at test time, which the others and ours do not. These methods require performances aligned to notehead coordinates, an annotation costly enough that nearly all training data is synthesized, so they learn on rendered audio and lose much of their accuracy in a real room, CUNet falling from 85.5% of onsets within 0.5 s to 22.4%, CYOLO-SB from 89.3% to 79.9% and CODA from 95.5% to 88.3%. Every remedy has been a perception remedy, whether reverberant augmentation [20], extra proprietary recordings [21] or a new architecture [22].

An audio-conditioned detector does not output a position but a scored set of candidate noteheads, of which the most confident is reported and the rest discarded. What that set contains bounds what any selection rule could achieve, and on the room-microphone recordings the bound lies far above what is realised. Among the 256 highest-scoring candidates



Fig. 1. One instant of a real room-microphone recording. In yellow the 64 most confident candidates. The most confident (red) is on the wrong staff.

at each instant, one falls within the 0.5 s tolerance at every onset, and a causal oracle, choosing with knowledge of the truth but forced to commit at each instant and never select a position behind its last, reaches 99.9%, against 80.0% for the deployed argmax (Fig. 1). That bound is not an achievable accuracy, but it leaves only a small irreducible error at the 0.5 s tolerance, which places most of the loss on real audio in candidate ordering rather than the visual evidence.

This argues for treating the task as a sequential decision over the positions a page affords rather than the regression of a coordinate. CANDOR decides from a transition model, scoring a displacement against elapsed time, and a set-based scorer, judging each candidate against its competitors in the same instant, trained listwise on synthesized audio alone and decoded causally and greedily at 1.25 ms per instant on one CPU thread. Thus we relocate the synthetic-to-real gap from perception to the decision and act on it, setting a new best on real recordings across three released detectors.

2. METHOD

2.1. Backbone

Every sheet-image follower shares a perception module, a convolutional network over the score page whose feature maps are modulated by the audio. The mechanism is feature-wise linear modulation [23], which injects a vector into a network by scaling and shifting its activations, $f_{\text{FiLM}}(x) = s(z) \odot x + t(z)$, where s and t are learned linear maps of a conditioning vector z and $\odot$ is elementwise. As the page does not indicate where the performer is, z carries the recent audio and the visual features become a function of what has just been heard.

CYOLO [20] encodes the last two seconds of audio into z with a spectrogram network and places FiLM layers through the downscale and upscale blocks of a YOLO detector [24], which then predicts a bounding box for the notehead currently sounding. CYOLO-SB [21] keeps that arrangement and adds heads for the enclosing system and bar. CODA [22] conditions its own backbone the same way, at the deeper stages, before cascading selection over layout regions supplied with the score. We use the released CYOLO detectors to propose noteheads.

2.2. CANDOR

CANDOR, for candidate ordering, sits after the backbone and replaces only its final step. At each instant the detector offers a few hundred possible notehead positions, and where it would report the most confident of them, CANDOR asks which one is most compatible with where its own previous decision was and with what the rest of the set looks like. The two parts below formalise these criteria, and each requires state to be carried across instants.

2.2.1. Candidate positions

A performance is processed in instants $t = 1, \dots, T$, and at each one the backbone reads the audio so far and the current page and scores note boxes $b_i = (x_i, y_i, w_i, h_i)$ with confidence o_i ; we keep the $K = 256$ most confident, $\mathcal{C}_t$. Staves are concatenated along one horizontal axis, the unrolled coordinate u_i , so positions across a line break are adjacent. The released detectors report the most confident box, $\hat{u}_t = \arg \max_i o_i$, the degenerate selector. CANDOR reads $\mathcal{C}_t$ and its previous choice $\hat{u}_{t-1}$, and nothing else. K is set where availability stops binding, a correct candidate being present at every onset only from $K = 256$. CANDOR saturates earlier, reaching 89.8%, 92.2%, 94.1%, 95.0% and 94.4% at $K = 16, 32, 64, 128$ and 256, the last two within the seed deviation.

2.2.2. Transition model

Between instants Δ_t frames apart the performer advances by a roughly predictable amount, so the displacement $d_i = u_i - \hat{u}_{t-1}$ is scored by a floored Gaussian whose mean grows with elapsed time,

$$\log \pi_i = \max \left(-\frac{(d_i - \mu \rho_t)^2}{2\sigma^2}, J \right), \quad (1)$$

with $\rho_t = \text{clip}(\Delta_t/\Delta_0, 0.2, 8)$ and $\Delta_0 = 5$ frames, the mean inter-onset gap in training. The floor J keeps a jump possible at bounded cost, and combined with confidence $\ell_i = \log o_i + \log \pi_i$ already recovers a large part of the gap.

2.2.3. Set-based scorer

The transition model scores each candidate independently given the history and therefore cannot express a comparative judgement. What it omits is whether a given box, relative to the others available at the same instant, is a plausible notehead rather than a spurious response. That comparison matters on real audio, where reverberation smears the onset the conditioning vector is built from, so the detector answers with many moderately confident boxes rather than one sharp response and absolute confidence is poorly calibrated. Each candidate is described by $\phi_i \in \mathbb{R}^{33}$, holding five confidence and rank terms within the instant, two box dimensions, seven

for motion from the previous decision, four for a constant-velocity extrapolation, six locating it in the coarse system and bar regions the backbone emits, and nine for how crowded and how typical it is among its neighbours, together with f_i , the 128-dimensional FiLM-conditioned feature the detection head scored the box from. A permutation-invariant network [25], as in learned re-scoring of detections [26], embeds each candidate as $e_i = g([\phi_i; \text{ReLU}(Wf_i)])$ and scores it against a context pooled over the set, $s_i = r([e_i; c])$ with $c = [\frac{1}{K} \sum_j e_j; \max_j e_j]$, the maximum taken per dimension. Here g and r are two-layer perceptrons and $W \in \mathbb{R}^{64 \times 128}$, so every candidate is judged against its competitors.

2.2.4. Training and inference

The decision is supervised listwise against soft targets, so with ε_i the difference in frames between the true onset time and the time the score maps candidate i 's position to, the quantity the evaluation thresholds, the target is $q_i \propto \exp(-\varepsilon_i/\tau)$ and the loss is the cross-entropy $-\sum_i q_i \log \text{softmax}(s)_i$, under which the scorer learns to order a set rather than regress a coordinate. History features come from previous decisions, perturbed by Laplace noise whose heavy tail exposes training to drift and gross error [27]. Target width and perturbation are chosen by ablation on the validation set: $\tau = 3$ frames, about 0.15 s, rewards landing near the onset rather than exactly on it, and noise of scale 30 px on 30% of samples is worth 2.9 points over an unperturbed history, which leaves the scorer below the argmax baseline it replaces; heavier noise is worse. At inference CANDOR decides

$$\hat{i}_t = \arg \max_{i \in \mathcal{C}_t} \beta s_i + (1 - \beta) \ell_i, \quad (2)$$

causally and greedily, one pass over 256 candidates per instant, in 1.25 ms on one CPU thread against the 50 ms available at 20 fps. Training on the residual $s_i - \ell_i$ and deciding $\arg \max_i (s_i + \ell_i)$ is Eq. 2 at $\beta = 0.5$, eliminating all four constants.

3. EXPERIMENTAL SETUP

3.1. Dataset

All experiments use MSMD [4], which pairs LilyPond engravings with MIDI-derived alignments for solo piano. Its repertoire is Western classical, dominated by Bach, with Beethoven, Mozart, Schumann and Chopin making up most of the remainder, and textures from two-voice inventions to dense Romantic writing, which sets how many plausible candidates a page affords at each instant. Every result below is therefore solo piano, the only repertoire with aligned sheet-image data at this scale.

The decision model is trained on 353 pieces, synthesized and convolved with measured room impulse responses [28], and validated on 19, using AdamW ($3 \cdot 10^{-3}$, cosine, 40 epochs,

Table 1. Real room-microphone recordings. $\dagger$ proprietary training data, $\ddagger$ layout annotation at test time.

Method	Onsets tracked (%) within threshold (s)					Gain
	≤ 0.05	≤ 0.1	≤ 0.5	≤ 1.0	≤ 5.0	
MM-Loc [16]	36.4	40.6	58.5	61.1	73.5	
RL agent [18]	18.5	20.0	47.6	60.3	90.1	
CUNet [19]	11.3	12.5	22.4	26.6	44.3	
CYOLO [20]	56.3	58.1	71.2	74.9	91.9	
CYOLO-SB [21]	61.0	63.0	79.9	83.2	96.0	
CYOLO-SB+A † [21]	68.2	70.6	86.5	89.1	98.1	
CODA ‡ [22]	72.1	74.3	88.3	91.4	98.7	
CANDOR _{CYOLO}	72.6	74.7	91.6	95.0	99.1	+20.4
CANDOR _{CYOLO-SB}	69.0	71.4	90.6	94.2	98.2	+10.7
CANDOR _{CYOLO-SB+A†}	76.0	78.5	94.4	96.9	99.6	+7.9
<i>Detectors we trained</i>						
CANDOR _{CYOLO-LSTM} (control)	58.2	60.6	83.3	87.5	98.3	+11.6
CANDOR _{CYOLO-Mamba}	62.8	65.7	91.5	94.6	99.3	+19.0
CANDOR _{CYOLO} , CODA tower [22]	44.7	47.4	76.2	83.9	96.5	+17.3

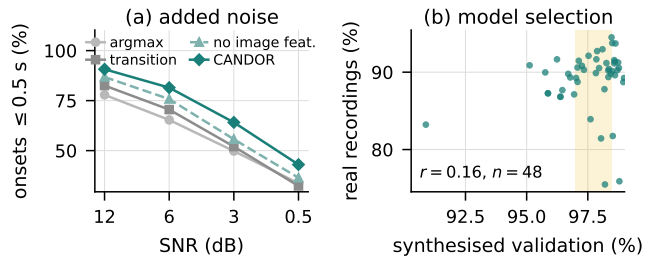


Fig. 2. (a) 80 held-out pieces under a room impulse response and additive noise, 40,594 onsets per level. (b) 48 trained decision models; half fall in the shaded 1.5-point validation band and differ by 19 points on real audio.

batch 256). Evaluation is the real-performance benchmark of [18], 16 pieces of Bach, Schumann, Mozart, Chopin and Mussorgsky spanning 25 pages and 4,149 onsets, played on a Yamaha AvantGrand N2 and captured with a room microphone, with the correct page given throughout. A further 80 held-out pieces are rendered with a real impulse response and noise from 12 to 0.5 dB SNR. We report the percentage of onsets whose predicted position maps to within 0.05 to 5 s of the true time, 0.5 s primary, with 95% intervals bootstrapped over pieces [29] and seed means with deviations.

The protocol separates three kinds of data across three stages, in that the decision model is trained on synthesized audio alone and never sees a real recording, the four constants (μ, σ, J, β) are calibrated leave-one-piece-out on the other 15 real pieces weighted equally, and evaluation is on the piece held out. That still lets the benchmark as a whole participate in choosing them, which the published baselines did not require, so the residual form of Eq. 2 is carried alongside as the variant fitting nothing to any real recording.

Table 2. Components of CANDOR on the real recordings, onsets tracked (%) at 0.5 s.

	n	argmax	+transition	full
<i>Perception model</i>				
CYOLO [20]	3	71.3	72.5	91.6 ± 1.4
CYOLO-SB [21]	6	80.0	86.5	90.6 ± 0.9
CYOLO-SB+A [†]	3	86.5	90.8	94.4 ± 0.9
<i>Decision model</i>				
residual, no fitted constants	3			88.7 ± 0.9
without temporal history	3			83.7 ± 0.4

4. RESULTS

With a detector $D(x_t, \mathcal{S}) \mapsto \mathcal{C}_t$, CANDOR is a causal selector $f(\mathcal{C}_t, \hat{u}_{t-1}) \mapsto \hat{u}_t$ and so a post-hoc decision layer for any follower exposing candidate locations, confidences and the previous decision, features optional. CODA [22] is not such a target, since its final stage regresses a continuous bar-local coordinate rather than choosing among proposals, and the sets it does select over, systems and bars, are supplied by layout annotation and already decoded with beam search.

Table 1 sets CANDOR against the published literature on recordings played straight through, the last column giving the improvement at 0.5 s over the perception model each row is built on. CANDOR tracks 94.4% of onsets within 0.5 s, 95% CI over pieces [92.2, 97.1]%, against a previous best of 88.3%, leading at every threshold. We acknowledge that sixteen pieces are few and the interval correspondingly wide, though the gain over the argmax it replaces, paired on the same resample, is +7.9 points with interval [+2.8, +12.2]. The margin does not depend on one perception model either, since retraining the decision on whichever candidates each detector emits gains 20.4, 10.7 and 7.9 points, and on the weakest, whose training data is entirely public, it passes where the strongest begins. On 94 synthetic test pieces it reaches 95.5% against 89.3% for argmax, and on the 80 noisy held-out pieces it beats the transition model at every level from 12 to 0.5 dB (Fig. 2a).

Since a selection rule is only as precise as the boxes it chooses between, the gain concentrates at the tolerances that measure which notehead is active rather than the position within it, and at 0.05 and 0.1 s CANDOR reaches 69.0% and 71.4% on CYOLO-SB candidates against CODA’s 72.1% and 74.3%. Table 2 decomposes the gain over the same candidates by adding Eq. 1 to the argmax and then the scorer, with n the number of decision-model seeds at a fixed detector, and shows that the transition model alone, with no trained weights, accounts for roughly half of it.

4.1. Ablation study

Giving the scorer the FiLM-conditioned backbone feature, DI-NOv2 patch features [30] on the same grid, or no image feature at all yields 90.6% over six seeds and 90.9% for each of two single runs, a spread well inside the 0.9-point seed deviation,

where on synthesized validation audio the same three differ by 3.4 points. Removing the features computed against the previous decision instead costs 7.2 points, the largest single ablation here, and teacher forcing attributes two thirds of the remaining error to drift, since given the true previous position the same scorer reaches 98.0% against the causal oracle’s 99.9%.

Retraining the detector from scratch with a different audio encoder gives the lower block of Table 1, not comparable with the converged checkpoints above it and reported because the decision gains a similar margin on any of them. A frozen MERT-95M tower [31] reaches 46.2%, driving training loss down by a factor of 4.7 while validation loss stays flat; the two-layer causal Mamba [32] of [22] in place of the whole audio tower reaches 76.2% against 83.3% for an identically budgeted control; and keeping the convolutional front end while replacing only the recurrence, which we call CYOLO-Mamba, reaches 91.5%. The recurrence therefore benefits from a state-space model while the convolutional front end is load-bearing, the encoder ablation absent from [22]. None of this rescues the gap, the strongest merely reaching what re-deciding over a released detector achieves.

Across 48 trained models, accuracy on the synthesized validation split correlates $r = 0.16$ with accuracy on the real recordings (Fig. 2b), and the best-validation model scores 89.3%, below one reading no image features at all. A reverberant held-out set correlates more strongly ($r = 0.80$) yet still selects poorly. Since the field selects models on synthesized audio, this is a caution beyond our own experiments.

5. CONCLUSION

The synthetic-to-real gap in sheet-image score following has been read as a failure of perception and answered with better audio, more data and new architectures. The measurement reported here places it elsewhere. On real room-microphone recordings the detector already proposes a correct position at every scored onset, and under the same realtime commitments the candidate set leaves only a small irreducible error, so at the standard tolerance most of what fails is the rule that reads a position from the evidence, not the evidence itself.

Acting on that, CANDOR orders the candidates the detector already emits and takes real piano recordings from 80.0% to 94.4% of onsets within 0.5 s, against 88.3% for the best published system, at 1.25 ms per instant on one CPU thread, without retraining any detector or reading layout annotation, across three released detectors. The decision depends far more on its temporal history than on the image representations we tested. Two results temper this, since accuracy on synthesized audio barely predicts accuracy on real audio across 48 trained models, which makes selection on a synthetic proxy unreliable for the field, not only for us, and since the forward-motion assumption of Eq. 1 confines the method to performances played in the written order. Pitch agreement is the next source of evidence independent of the history.

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